

Qiqi Hu

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RESEARCH INTERESTS

AI for energy storage and thermal safety; data-driven modeling for battery thermal runaway; scientific machine learning; multiphysics simulation; robust optimization and trustworthy AI for energy systems.

PROFILE

Master's student in Environmental Science and New Energy Technology at Tsinghua University (expected Dec. 2026), with a unique interdisciplinary background spanning cybersecurity, energy engineering, and artificial intelligence. My research lies at the intersection of AI and energy storage, focusing on battery thermal safety, interpretable machine learning, and multiphysics modeling. For my Ph.D. studies, I aim to develop data-driven, interpretable, and robust AI frameworks for safety-critical energy systems, accelerating innovation in battery safety control and sustainable energy optimization.

EDUCATION

Tsinghua University <i>M.S. in Environmental Science and New Energy Technology</i> <i>Advisor: Prof. Hongda Du</i>	2023 – 2026
Southern University of Science and Technology <i>Visiting Student, Department of Computer Science and Engineering</i>	2024 – 2025
Qingdao University <i>B.S. in Information Security, School of Computer Science and Technology</i>	2019 – 2023

RESEARCH EXPERIENCE

Postgraduate Researcher, Tsinghua University • Guangdong Provincial Key Laboratory of Thermal Management Engineering and Materials. • Conduct research on battery thermal management, thermal runaway safety, and AI-driven modeling for energy systems. • Developed interpretable machine learning methods for pre-event severity prediction of lithium-ion battery thermal runaway using battery design metadata. • Performed thermal modeling, multiphysics analysis, and data-driven optimization for battery thermal management systems under high-power operating conditions. • Advisor: Prof. Hongda Du	Sep. 2023 – Present
Visiting Student, Southern University of Science and Technology • Visual Intelligence and Perception Lab, Department of Computer Science and Engineering. • Conducted research on trustworthy diffusion models and copyright/content security for AIGC systems. • Proposed a protection framework integrating digital watermarking and latent-space perturbation for diffusion-based image-to-image generation. • Advisor: Prof. Feng Zheng	Mar. 2024 – Dec. 2024
Research Intern • Shandong Provincial Undergraduate Innovation and Entrepreneurship Training Program (Grant No. S202111065050). • Studied privacy-preserving computation and secure outsourced clustering for resource-constrained IoT scenarios. • Advisor: Prof. Hanlin Zhang	Dec. 2021 – Dec. 2022

SELECTED PUBLICATIONS

[J2] Safety-Oriented Pre-Event Severity Prediction of Lithium-Ion Battery Thermal Runaway Using Interpretable Machine Learning [Code](#)

First author, accepted by [Process Safety and Environmental Protection](#) (IF: 7.8)

- Proposed a metadata-driven machine learning framework for predicting thermal runaway severity without requiring calorimetry or ejection experiments.
- Developed an interpretable Severity Index and a CatBoost-based classifier, achieving 81% accuracy and 100% recall for high-severity cells.
- Used SHAP-based interpretation and sensitivity analysis to identify key design factors and evaluate engineering robustness.

[J1] Design and Multi-Objective Optimization of an Efficient UAV Battery Thermal Management System Using a PCM–Air Synergistic Cooling Strategy

First author, accepted by [Applied Thermal Engineering](#) (IF: 6.9)

- Designed a hybrid battery thermal management system combining phase change materials and air cooling for high-power UAV operating conditions.
- Conducted multiphysics simulations and parametric analysis, followed by entropy-weighted TOPSIS for multi-objective optimization.
- Reduced maximum battery temperature by 43.3% to below 46.8 °C, with only a 22.2 wt% increase in system mass.

SELECTED RESEARCH PROJECTS

[P2] Copyright Protection for Diffusion-Based Image-to-Image Generation [Code](#)

- Proposed a dual-protection framework that combines digital watermarking and latent-space adversarial perturbations for diffusion models.
- Achieved robust and transferable copyright protection without model fine-tuning while preserving image quality and reliable watermark extraction.

[P1] Privacy-Preserving Shared Nearest Neighbor Clustering for the Internet of Things [Preprint](#)

- Developed a cloud-assisted privacy-preserving SNN clustering framework for resource-constrained IoT devices using orthogonal matrix transformation.
- Reduced local computation time from 10.859 s to 1.183 s while preserving clustering performance on real-world datasets.

HONORS AND AWARDS

- **Typical Graduate (Ranked 1st)**, School of Computer Science and Technology, Qingdao University, 2023
- **Qingdao University Star Award**, highest university-wide student honor, 2023
- Honorable Mention, International Mathematical Contest in Modeling (MCM), 2021

TECHNICAL SKILLS

Programming	Python, MATLAB, C, $\text{\LaTeX}$
ML Libraries	PyTorch, TensorFlow, scikit-learn
Simulation / Tools	COMSOL, Linux, VS Code, Anaconda
Languages	Chinese (native), English (fluent)